School of Science, Computing and Emerging Technologies

COS80023 Big Data

Pass Task 5: Information Retrieval with Pig

Overview

Learn how to use Pig for information extraction on Azure HDInsight.

Purpose

Demonstrate competency at extracting information from flat files using Pig.

Task

Carry out the tasks described below and answer the questions in your submission.

Time

This task should be completed in the fifth lab class or the week after and submitted to Canvas for feedback. It should be discussed and signed off in tutorial 6 or 7.

This task should take no more than 2 – 2 ½ hours to complete.

Resources

- Presentation (from Canvas)
- SalesJan2009.csv file from Canvas
- MS Azure tutorial for the creation of clusters: https://docs.microsoft.com/en-us/azure/hdinsight/hadoop/apache-hadoop-linux-tutorial-get-started (If you use the link to quickstart on this page, you will not see all the options discussed in the task)
- Any other online material
- genAI – Allowed for research. You must formulate the answers in your own words and be able to answer questions about the topics

Feedback

Discuss your answers with the tutorial instructor.

Next

Get started on module 6.

Pass Task 5 — Submission Details and Assessment Criteria

Write down the questions and answers in a text or Word document, convert to pdf and upload to Canvas. Your tutor will mark the submission on line. If the submission is not marked as '1', it is considered as incomplete and must be resubmitted.

Task 5

Run Pig Latin commands on an Azure shell. Pig is a part of the Hadoop distribution and you have access to it when you create a Hadoop cluster using HDInsight.

Overview of steps.

1. Create a Hadoop cluster using HDInsight as well as Azure Storage account.
2. Upload a file to Azure Storage using the portal.
4. Through terminal connect to your cluster using ssh.
5. Run the commands provided against the file you uploaded.
6. Change the commands to extract different information.
7. Repeat the wordcount example in Pig.

1. Deciding your location

Our subscription has access to several locations, but the resources in each location are limited. As we all work at the same time, please choose your own location as follows:

Student number is odd (ends in a 1, 3, 5 etc.): <yourlocation> = Australia East

Student number is even (ends in 2, 4, 6 etc.): <yourlocation> = Australia Southeast

It is fine to have your resource group in a different location, but all resources (storage, clusters) should be chosen from your location to ensure nothing goes wrong.

2. Creating a HDInsight Cluster for Hadoop

As you know, in Hadoop tasks run on a cluster of nodes. First, you have to create the cluster.

Assuming you are already logged in, go to your Home. In the search field top center of the page, type HDInsight. Choose HDInsight clusters from the options.

Click on '+Create'.

Select correct subscription (containing COS80023) and your resource group.

Set the cluster name to s<yourstudentnumber>cluster (no upper case letters allowed), e.g. s12345678cluster.

Choose <yourlocation> as location.

Choose Hadoop 3.1 as cluster type. Leave the default cluster username and ssh user. Choose a password with upper case, lower case, numbers and a special character. Make a note of the password, you will need it later.

Click Next to proceed to Storage. Select Azure Storage. Click Create new. Name your new storage <yourstudentnumber>storage.

For the container choose <yourstudentnumber>container.

Leave the other options as default.

Click Next to proceed to Security and Networking. Do not change the default options.

Click Next to proceed to Configuration+pricing. Examine the default resources for the cluster. There are Head nodes, Zookeeper nodes and Worker nodes.

Choosing the smallest possible options might lead to using resources that are scarce in Australian locations. Please change the Head node to E2 V3. Leave the other two, but reduce the Worker node to 1 instead of 4 (observe the pricing that adjusts with the choice).

Node type	Node size	Number of ...	Estimated cost/h...
Head node	E2 V3 (2 Cores, 16 GB RAM), 0.25 AUD/hour	2	0.50 AUD
Zookeeper node	A2 v2 (2 Cores, 4 GB RAM), 0.00 AUD/hour (F...	3	0.00 (FREE)
Worker node	E8 V3 (8 Cores, 64 GB RAM), 1.01 AUD/hour	1	1.01 AUD

☐ Enable managed disk

No need to add script action.

Click Review+create. On the summary page, you get to create the cluster. It typically takes several minutes for the cluster to deploy.

To find out about the progress (and possible errors), click on notifications on the top right (bell-shaped icon).

1. Uploading the file in storage.

In your Dashboard, go to Storage Accounts.

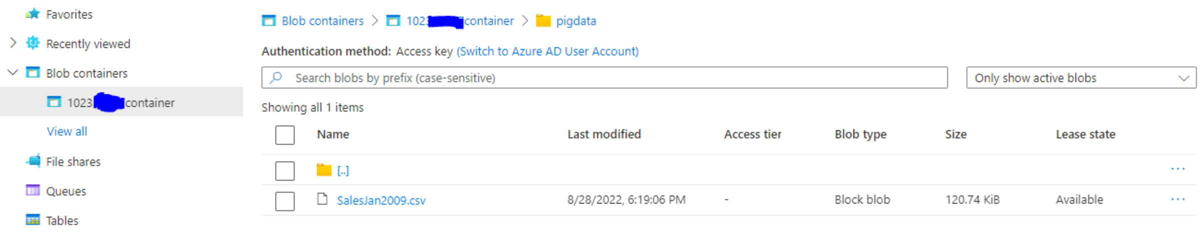
You should see the storage account that you have created. If you can't, try selecting the correct subscription. Click on your storage account.

In the left-hand side of the page you should see "Storage Browser", click on it.

Click on your blob container to see the folders inside.

Create a new folder by clicking +Add Directory, name your new folder pigdata and press ok. This will take you inside pigdata folder. You need to upload the file SalesJan2009.csv now. You can find this file on Canvas, under resources for this task. If you do not upload the file and go back, then creation of pigdata folder will not materialize.

Your Storage Explorer should look similar to this:



3. Connect to cluster using ssh

Open terminal and type the command for an ssh connection:

ssh sshuser@<yourstudentnumber>cluster-ssh.azurehdinsight.net

Alternatively, when your cluster is open in your dashboard, you can click on the item 'SSH+Cluster login' to be able to copy the connect string.

Warning: If you create the cluster the second time (after deleting it the first time), the signature will have changed. The prompt will no longer ask you, but tell you connecting is too dangerous:

```
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!     @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ECDSA key sent by the remote host is
SHA256:MTM6ADJYDK6GXxUFjvOclNjBt16f7hHkFPqUif8MN8I.
Please contact your system administrator.
Add correct host key in C:\\Users\\Me/.ssh/known_hosts to get rid of this message.
Offending ECDSA key in C:\\Users\\Me/.ssh/known_hosts:1
ECDSA host key for cos80023cluster-ssh.azurehdinsight.net has changed and you have requested strict checking.
Host key verification failed.
```

You have to find the known_hosts file and erase the entry for this connection (asurehdinsight.net) before you can continue.

Once you see the command prompt for the sshuser, you can type

```
pig
```

and Enter. If your command prompt reads grunt>, you are connected to Pig.

4. Run the following commands and find out what they do

First, inspect the content of the SalesJan2009.csv file and find out what it represents.

Run the following line:

```
rawsales = LOAD '/pigdata/SalesJan2009.csv' USING PigStorage(',') AS
(Transaction_date:chararray, Product:chararray, Price:int,
Payment_type:chararray, Name:chararray, City:chararray, State:chararray,
Country:chararray);
```

Question 1: What does each part of this line do?

It may help to visualise the outcome of the command by typing

```
DUMP rawsales;
```

The following command strips the headers (first line) from the table:

```
salesdata = FILTER rawsales BY Price IS NOT NULL;
```

Question 2: What does each line do (explain every detail)?

```
salesandcountry = FOREACH salesdata GENERATE (chararray) Country AS
country, (int) Price AS price;

countrygroups = GROUP salesandcountry BY country;

salespercountry = FOREACH countrygroups GENERATE group as country,
SUM(salesandcountry.price) AS totalsales;
```

Question 3: What do these two lines do?

```
sortedsales = ORDER salespercountry BY totalsales desc;
STORE sortedsales INTO '/pigdata/SalesByCountry' USING PigStorage(' ');
```

5. Adjust the commands to extract the sales per credit card

Use the same data to extract how the sales transactions distribute over the types of credit card.

Run the code in Pig using the terminal. Store the answer in a different directory in pigdata (not SalesByCountry but some other name you choose).

Question 4: – Document your code in your answer document.

Take a screenshot of the content of the new folder you created in pigdata and add to your answer document.

6. Repeat the wordcount example from last week using Pig

The code for this is not provided. There are plenty of examples on line. Run them using your file you used for MapReduce (or make a new one if you have lost it).

Explain the steps and provide a screenshot of the content of the result (file or variable).

Important: You MUST delete the cluster at the end. The cluster will keep running unless you delete it. On your dashboard, go to All resources. Tick the box in front of your cluster and click Delete. You have to confirm in the next step.

Observe the notifications (bell icon). When the cluster has been deleted, refresh and put a screenshot of the resources without the cluster in your answer document.